A FLAT MODULE OF FINITE PRESENTATION IS LOCALLY A DIRECT SUMMAND OF A FINITE FREE MODULE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The local presentation of a module of finite presentation by a map between finite free modules followed by a surjection. Used at the start of the counted proof (source lines 3161-3166). Provides the two free modules whose rank controls the number of steps of the iteration.
- The local factorisation criterion for a map followed by a surjection onto a flat module. Invoked inside the counted proof at source lines 3167-3177 and attributed to a lemma of this chapter. Allows the proof to peel off one summand of the relations at a time.

Source reference. Original lemma lemma-flat-eq: sources/stacks-modules/pinned-original-source.tex, source lines 3167–3167. Navigation pointer only: the counted text refers to the Stacks item lemma-flat-eq, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a flat $\mathcal{O}_X$-module of finite presentation. Then $\mathcal{F}$ is locally a direct summand of a finite free $\mathcal{O}_X$-module.*

Proof. After replacing X by the members of an open covering, we may assume there exists a presentation

$$\mathcal{O}_X^{\oplus r} \to \mathcal{O}_X^{\oplus n} \to \mathcal{F} \to 0$$

Let $x \in X$. By Lemma lemma-flat-eq we can, after shrinking X to an open neighbourhood of x, assume there exists a factorization

$$\mathcal{O}_X^{\oplus n} \to \mathcal{O}_X^{\oplus n_1} \to \mathcal{F}$$

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such that the composition $\mathcal{O}_X^{\oplus r} \to \mathcal{O}_X^{\oplus n} \to \mathcal{O}_X^{\oplus n_1}$ annihilates the first summand of $\mathcal{O}_X^{\oplus r}$. Repeating this argument $r - 1$ more times we obtain a factorization

$$\mathcal{O}_X^{\oplus n} \to \mathcal{O}_X^{\oplus n_r} \to \mathcal{F}$$

such that the composition $\mathcal{O}_X^{\oplus r} \to \mathcal{O}_X^{\oplus n} \to \mathcal{O}_X^{\oplus n_r}$ is zero. This means that the surjection $\mathcal{O}_X^{\oplus n_r} \to \mathcal{F}$ has a section and we win. $\square$